

Profile

As a professional software engineer, I have technical experience across a variety of domains, including front-end and back-end development, data engineering, data analysis, DevOps, and cloud architecture. I have worked on dynamic, cross-functional teams responsible for designing and developing new applications and supporting legacy systems. I am seeking opportunities where I can apply my functional expertise, drive business impact, and grow as a technical leader.

Education

- University of Virginia, Charlottesville, VA Aug 2014 – May 2018
 - Bachelor of Arts: Mathematics and Computer Science
- University of Washington, Seattle, WA Jan 2026 – Present
 - Professional Master's Certificate: Applied Cybersecurity Engineering

Work and Research Experience

- Senior Software Engineer | CarMax, Inc. Jun 2021 – Present
(Software Engineer II: Jun 2021 – Aug 2023)

- Designed and developed scalable C#.NET microservices used throughout the organization to securely manage 500,000 customer documents per month in support of critical business operations
- Developed customer-facing and associate-facing full-stack web applications (C#.NET, TypeScript, React) used to upload, review, and transmit documents, reducing manual intervention on 100,000 transactions per month
- Partnered with architects, engineers, and product managers to define a roadmap for re-platforming legacy applications, improving reliability, security, maintainability, and developer efficiency
- Collaborated across product teams to define data contracts, establish SLAs, and triage incidents
- Designed and developed an Azure Data Factory pipeline to synchronize data across systems
- Developed a prototype using Azure AI Foundry and Document Intelligence to automate document review
- Published and consumed enterprise events to facilitate highly asynchronous business processes
- Developed and promoted reusable NuGet packages for shared data models and gateway connections
- Owned end-to-end software lifecycle, including CI/CD pipelines, Azure infrastructure, and production monitoring

- Freelance Software Engineer | FlyCast River Forecast, LLC Mar 2021 – Oct 2023

- Developed a full-stack Python/Django web application to register customers to a membership rewards program
- Developed a Python/Flask API and static React web application to help anglers improve their fly tying skills

- Senior Consultant | CapTech Ventures, Inc. Aug 2018 – May 2021
(Consultant: Aug 2018 – Jan 2021)

Fortune 500 insurance holding company (Apr 2020 – May 2021)

- Developed a full-stack Python/Django web application for adjusters to track claims within an interactive dashboard
- Enhanced existing SQL database ETL processes to normalize data originating from disparate source systems

Fortune 500 automotive retailer (Feb 2019 – Apr 2020)

- Developed a full-stack C#.NET and React web application to replace a legacy sales application
- Used Azure SignalR to relay financing offers from lenders to customers in real time
- Wrote automated, full end-to-end integration tests using WebdriverIO
- Designed a production support dashboard for monitoring application usage using Azure App Insights

Fortune 500 manufacturing company (Sep 2018 – Dec 2018)

- Developed a .NET API for verifying user identities by integrating with a third-party provider (LexisNexis)

- Data Analyst Intern | IST Research Corp. May 2017 – Aug 2017

- Developed a Python/Flask API to perform reverse image searches and natural language processing tasks, such as sentiment analysis and named entity recognition, by leveraging an Elasticsearch instance

- Undergraduate Teaching Assistant | UVA Computer Science Department Jan 2017 – May 2018

- Supported students in developing formal proofs and understanding set theory over three semesters, under three different instructors, for the course Discrete Mathematics

- Undergraduate Researcher | UVA Mathematics Department May 2016 – Aug 2016

- Partnered with the department chair to study hypergraphs with specific characteristics and used a mathematical programming language (Macaulay2) to generate data supporting an unproven conjecture in abstract algebra

Skills and Technologies

- Programming languages: C#, Java, JavaScript, TypeScript, Python, SQL
- Tools and frameworks: .NET, Entity Framework, Node, Express, React, Webpack, Vite
- Cloud services: Azure and Amazon Web Services (AWS)